Homework #3

In this homework, you need to write an ARM assembly program for matrix computation. The program is expected to execute on GDB ARM emulator for the verification.

Write an ARM assembly program to do the matrix computation.
(請參閱 `matrix1_2.s`，並做延伸)

- A is a 1x3 matrix and B is a 3x4 matrix
- C is a 1x4 matrix
- A, B, and C are stored in row-major layout. (similar to Figure 1)
- $C = A \times B$
- Figure 1 shows the layout of C.
- Each element in A, B, and C is a word-sized signed integer.
- Write an ARM assembly program to compute C.
- 程式碼必須加上適當的註解
- The integer values of A and B are assigned by yourself.
- The overflow/underflow problems are not considered during the computation.
- After computation, insert a NOP instruction. At this point in the program, register r0 will point to the address of C's first element.
- 請勿繳交【利用編譯器所自動產生的組合語言程式】
- 請勿抄襲
- 可以使用 **GPT** 協助你撰寫程式，但請一定要自己理解，並加上適當的註解

You need to turn in the following files to **eCourse2 system** (http://ecourse2.ccu.edu.tw/):

1. "README.txt or README.pdf" file describes the features in your program and how to compile and run your program.
2. Your ARM assembly program, named hw3.s, with proper comments.
3. Executable code: hw3.exe
4. Makefile (Makefile 中預設 arm-none-eabi-gcc 執行檔路徑已經加入 PATH 環境變數中)
5. Any other files are required in your implementation.
6. 請將欲繳交的檔案壓縮成 <hw3_學號.tar.xz>，上傳壓縮檔

Due Data: October 28 (Wednesday), 24:00, 2026

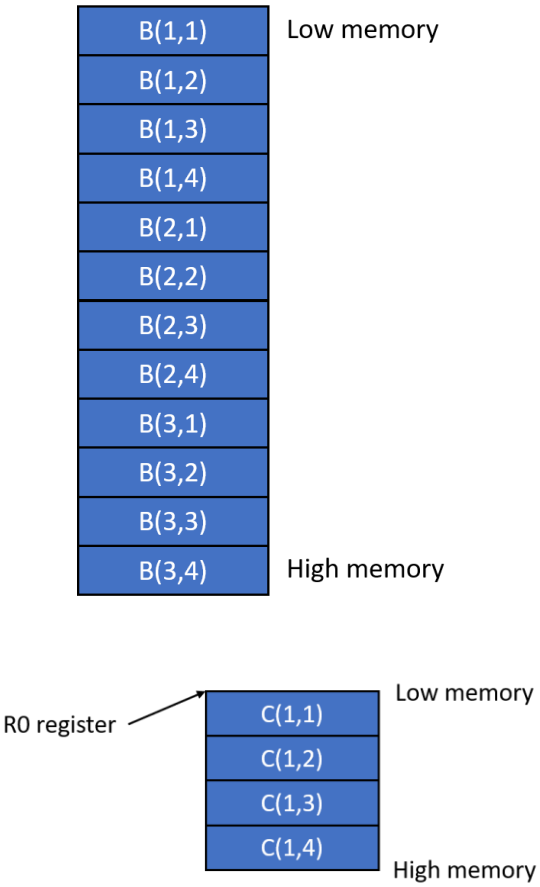


Figure 1: Layout of a matrix.